4

Classical dataset characteristics

The pioneering work by Adomavicius and Zhang (2012) first demonstrated the existence of a correlation between the intrinsic characteristics of recommender system datasets and the final performance achieved by learning algorithms. This finding opened the possibility of anticipating algorithmic behavior based on **dataset properties**, even before model training takes place. Before delving into the topological characteristics, which are the focus of the following section, we first introduce the so-called **classical dataset characteristics**, that is, those already well established in the literature and whose effectiveness has been consistently demonstrated across a variety of recommendation settings (Deldjoo *et al.*, 2021; Deldjoo *et al.*, 2020; Malitesta *et al.*, 2024).

4.1 Space size

The space size quantifies the total number of possible user–item interactions in a dataset. When comparing two datasets with the same number of observed interactions, the one with a larger space size exhibits a lower probability of observing a link between a given user and item. Consequently, the space size provides an estimate of the potential con-

centration of interactions among users and items. It is formally defined as:

$$\Psi = U * I, \tag{4.1}$$

where U and I denote the total number of users and items, respectively. The metric ranges from 0 (no possible interactions) to $+\infty$ as the dataset size grows.

4.2 Shape

While the space size captures the magnitude of the interaction space, it does not provide any information about the balance between the number of users and items. Indeed, two datasets with the same space size may differ substantially in their composition; for instance, one may have many users and few items, while the other exhibits the opposite pattern. This relationship is expressed by the shape, defined as:

$$\Phi = \frac{U}{I}. \tag{4.2}$$

A user–item ratio of 1 indicates a perfect balance between the number of users and items. Values close to 0 correspond to datasets with relatively few users and many items, whereas very high (potentially unbounded) values indicate the opposite configuration.

4.3 Density and sparsity

Data sparsity is one of the defining characteristics of recommendation datasets, as the recommendation task typically involves suggesting items from a large catalog with which each user has interacted only to a limited extent. A highly sparse dataset conveys less information than a denser one, given the same space size. In such sparse settings, the probability that two users (or items) share interactions with the same items (or users) decreases, which in turn weakens the effectiveness of collaborative filtering approaches.

To define data sparsity, it is first needed to introduce data density, which measures the proportion of existing interactions out of all possible ones:

$$\Delta = \frac{E}{U * I}, \tag{4.3}$$

where $E = |\{(u, i) \mid \mathbf{R}_{u,i} = 1\}|$ is the number of interactions existing among users and items in the recommendation data.

Since the denominator represents the total number of possible interactions, the density can equivalently be expressed as:

$$\Delta = \frac{R}{\Psi}, \tag{4.4}$$

where Ψ denotes the space size introduced in Equation 4.1. The density ranges from 0 (maximum sparsity) to 1 (maximum density). Accordingly, sparsity is defined as the complement of density:

$$\Lambda = 1 - \Delta. \tag{4.5}$$

4.4 Gini coefficient

The Gini coefficient measures the inequality in the frequency distribution of users or items. In practice, it quantifies how unevenly users and items are exposed within a dataset. This metric is particularly relevant in recommendation scenarios, where exposure is typically far from uniform: a small set of highly popular items often dominates the catalog, while the majority of users interact with only a limited subset of items.

The user- and item-level Gini coefficients are defined as follows:

$$\mathrm{K}_{\mathcal{U}} = \frac{\sum_{u=1}^{U-1} \sum_{v=u+1}^{U} |\sigma_u - \sigma_v|}{U \sum_{u=1}^{U} \sigma_u}, \qquad \mathrm{K}_{\mathcal{I}} = \frac{\sum_{i=1}^{I-1} \sum_{j=i+1}^{I} |\sigma_i - \sigma_j|}{I \sum_{i=1}^{I} \sigma_i}, \tag{4.6}$$

where U and I denote the number of users and items, respectively, and σ_k represents the interaction frequency of user u or item i. The operator $|\cdot|$ denotes the absolute value. A Gini coefficient close to 0 corresponds to a perfectly balanced distribution, whereas a value close to 1 indicates a highly unbalanced one.

4.5 Empirical evaluation of classical characteristics

To provide an indication of the typical ranges of these metrics and to highlight their variability across datasets, we compute all characteristics for the examples introduced in Table 4.1. The datasets are ordered by their number of interactions, from the largest (*MovieLens 1M*) to the smallest (*LastFM*).

The **space size** (Ψ) exhibits substantial variation, spanning more than two orders of magnitude, reflecting the different scales of the underlying user–item interaction spaces. As expected from its definition, this metric is positively related to density: larger interaction spaces tend to be associated with lower densities, particularly in highly sparse, real-world settings such as *Yelp* and Amazon-based datasets. Notably, large values of Ψ can also have practical implications for the size and complexity of recommendation models, especially for embedding-based architectures in which the number of users and items directly determines the dimensionality, and therefore the memory footprint, of the embedding tables.

The **shape metric** (Φ) also reveals domain-dependent structural differences. Values greater than one indicate that the number of users exceeds the number of items, a condition observed in most datasets, sometimes by a factor of two. *Amazon Books* constitutes an exception, with a shape below one, suggesting a catalog-heavy configuration. Such contrasts align with the intrinsic nature of each domain: e-commerce collections typically feature extensive item catalogs, whereas media- or location-oriented platforms tend to exhibit a more balanced or user-heavy structure.

The dataset **density** (Δ) is uniformly low, ranging from 0.003 to 0.060, and reflects the well-known sparsity characteristic of recommendation data. *MovieLens 1M* appears as an outlier with a substantially higher density, which contributes to its reputation as a comparatively "easy" benchmark and partly explains the strong performance typically reported on this dataset relative to more sparse alternatives (Anelli *et al.*, 2022a; Fan *et al.*, 2024).

The **Gini coefficients** ($K_{\mathcal{U}}$, $K_{\mathcal{I}}$) further illuminate differences in interaction distributions. Inequality levels are generally higher for items

Table 4.1: Calculated classical characteristics for our selected recommendation datasets.

Dataset	Ψ	Φ	Δ	$\mathbf{K}_{\mathcal{U}}$	$\mathbf{K}_{\mathcal{I}}$
MovieLens 1M	$1.65 \cdot 10^7$	1.914	0.060	0.507	0.548
Yelp	$1.83 \cdot 10^8$	1.177	0.004	0.341	0.376
Amazon Beauty	$9.70 \cdot 10^7$	1.327	0.018	0.254	0.332
Amazon Books	$7.95 \cdot 10^6$	0.732	0.003	0.408	0.304
Gowalla	$1.52 \cdot 10^6$	1.105	0.025	0.327	0.369
LastFM	$5.33 \cdot 10^6$	1.683	0.035	0.155	0.484

than for users, consistent with longstanding observations regarding popularity concentration. A notable case is *LastFM*, where the item-side Gini is substantially higher than the user-side value. This indicates that artists are far more unevenly exposed than users, with a small number of very popular artists attracting a disproportionate share of interactions. This behavior is characteristic of music consumption, where a limited set of artists typically dominates listening patterns across the platform.

Finally, several domain-level patterns emerge. In this sense, location-based (*Gowalla*) and music-oriented (*LastFM*) datasets tend to exhibit moderate densities and shape values above one. E-commerce datasets (*Amazon Beauty* and *Amazon Books*) are extremely sparse and display moderate inequality patterns. *Yelp* combines one of the largest interaction spaces with one of the lowest densities, underscoring the complexity of review-based platforms. In contrast, *MovieLens 1M* stands out as structurally more compact, denser, and more skewed, reinforcing its role as a convenient yet less representative benchmark for contemporary recommendation research.